

Playwright Testing

Documentation

Purpose of this document. A reference demo report that covers every front the gate asks for — playwright tests.

- Assessment: Playwright testing on realtime web application

Website URL: <https://www.nykaa.com/>

1. Objective & user story:

Automate the basic homepage to search page flow of Nykaa shopping journey on a maintainable run it headed mode for stable explicit waits. User story. As a shopper, I want to enter the home and check the page loaded and the banner confirmed.

Now check the visibility of search bar and type any product name to find the product. Verify the product exists or not. Verify for negative test

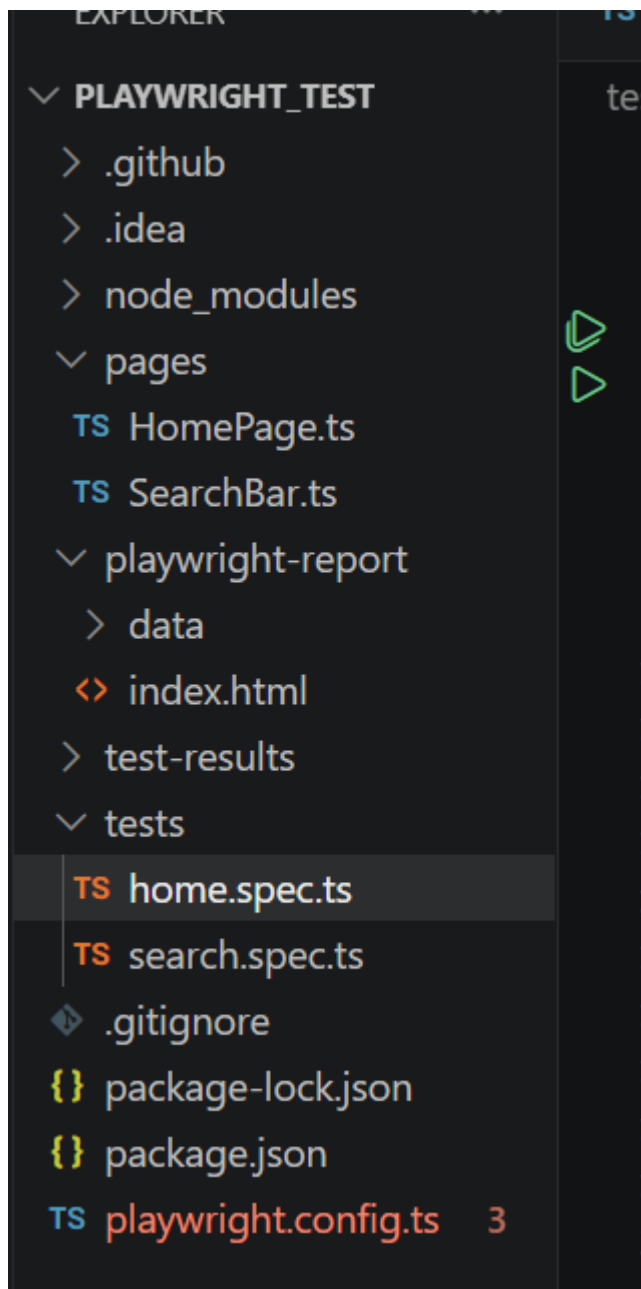
2. Environment & Setup:

Language: typescript

Environment: VS Code

Packages: Node modules

Project Structure



HomePage.ts:

```
TS HomePage.ts X TS home.spec.ts TS search.spec.ts TS playwright.config.ts 3 TS SearchBar.ts
pages > TS HomePage.ts > 🚀 HomePage > 📖 navigateToHomePage
1  import { expect, test } from '@playwright/test';
2  import { Page } from '@playwright/test';
3
4
5  export class HomePage {
6
7      constructor(private readonly page: Page){
8      }
9
10     async goto(){
11         await this.page.goto('https://www.nykaa.com/');
12     }
13
14     async navigateToHomePage() {
15         await this.page.goto('https://www.nykaa.com/');
16         await this.page.waitForLoadState('networkidle');
17         await this.page.locator('div').filter({ hasText: 'Nykaa' }).nth(3).click()
18         await this.page.locator('div').filter({ hasText: 'Nykaa' }).nth(2).click()
19     }
20
```

SearchPage.ts:

```
TS HomePage.ts TS home.spec.ts TS search.spec.ts TS playwright.config.ts 3 TS SearchBar.ts X
pages > TS SearchBar.ts > SearchBar > searchbarUnknownProductFound
1 import { expect, test } from '@playwright/test';
2 import { Page } from '@playwright/test';
3
4 export class SearchBar {
5     constructor(private readonly page: Page) {
6     }
7
8     async goto(){
9         await this.page.goto('https://www.nykaa.com/');
10    }
11
12    async searchbarFound(){
13        // await this.page.locator('.active.css-1nzkux7').click();
14        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).click();
15        await this.page.locator('#headerMenu').getByRole('button').click();
16    }
17
18    async searchbarProductFound(){
19        // await this.page.locator('.active.css-1nzkux7').click();
20        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).click();
21        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).fill('Shampoo');
22        await this.page.keyboard.press('Enter');
23    }
24
25    async searchbarUnknownProductFound(){
26        // await this.page.locator('.active.css-1nzkux7').click();
27        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).click();
28        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).fill('xyzz');
29        await this.page.keyboard.press('Enter');
30    }
31
32    async searchbarProductNotFound(){
33        // await this.page.locator('.active.css-1nzkux7').click();
34        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).click();
35        await this.page.getByRole('textbox', { name: 'Search on Nykaa' }).fill('@#$');
36        await this.page.keyboard.press('Enter');
37    }
38
39 }
40
41
```

Test Files: The test codes for home page testing and search bar testing

Home.spec.ts:

```

> TS home.spec.ts > test.describe("Home Page Tests") callback > test("should navigate to the home page") callback
import { test, expect } from '@playwright/test';
import { HomePage } from '../pages/HomePage';

test.describe('Home Page Tests', () => {
  test('should navigate to the home page', async ({ page }) => {
    const homePage = new HomePage(page);
    await homePage.goto();
    await expect(page).toHaveURL('https://www.nykaa.com/');
    // await expect(page).toHaveTitle(/Nykaa/);
    await page.getByRole('link', { name: 'Nykaa Logo' }).click();
    await expect(page).toHaveURL('https://www.nykaa.com/?root=logo');
    // await expect(page).toHaveTitle("Nykaa");
    await page.locator('.image-wrapper.css-19k52lt > .css-13o7eu2').click();
    await expect(page).toHaveURL('https://www.nykaa.com/luxe/brands/sk-ii/c/98172?transaction_id=061587');
  });
});

```

Search.spec.ts:

```

import { expect, test } from '@playwright/test';
import { SearchBar } from '../pages/SearchBar';

test.describe('Search Bar Tests', () => {
  test('should navigate to the home page and find search bar', async ({ page }) => {
    const searchBar = new SearchBar(page);
    await searchBar.goto();

    searchBar.searchbarFound();
    await expect(page).toHaveURL('https://www.nykaa.com/');
    await expect(page).toHaveTitle(/Nykaa/);
  });

  test('Positive: should enter a value in the searchbar and find a product', async ({ page }) => {
    const searchBar = new SearchBar(page);
    await searchBar.goto();
    searchBar.searchbarProductFound();
    await expect(page).toHaveURL('https://www.nykaa.com/hair-care/hair/shampoo/c/316?search_redirection=True');
    await expect(page.getByRole('heading', { name: 'Buy Shampoos Online, 1380' })).toBeVisible();

    searchBar.searchbarUnknownProductFound();
    await expect(page).toHaveURL('https://www.nykaa.com/search/result/?q=xyzz&root=search&searchType=Manual&sourcepage=Search+Page');
    await expect(page.getByRole('heading', { name: 'Showing 12 of 12 results for xyzz' })).toBeVisible();
  });

  test('Negative: should enter a value in the searchbar and not find a product', async ({ page }) => {
    const searchBar = new SearchBar(page);
    await searchBar.goto();
    await searchBar.searchbarProductNotFound();
    await expect(page).toHaveURL('https://www.nykaa.com/search/result/?q=%40%23%24&root=search&searchType=Manual&sourcepage=home');
    await expect(page.getByAltText('no_result')).toBeVisible();
    await expect(page.getByText('Thanks for visiting our')).toBeVisible();
    // await expect(page.locator("#title", { hasText: 'Thanks for visiting our website!' })).toBeVisible();
    await expect(page.getByText("Unfortunately, we couldn't find any matches for . Please try searching with another term.")).toBeVisible();
  });
});

```

How to run:

1. Install playwright

Cmd: npx init playwright@latest

3. Install node packages

4. Run command to execute tests:

Cmd: npx playwright test 'file name'

Test Results ScreenShot:

```
Running 4 tests using 1 worker
✓ 1 [chromium] › tests\home.spec.ts:5:7 › Home Page Tests › should navigate to the home page (8.4s)
✓ 2 [chromium] › tests\search.spec.ts:6:7 › Search Bar Tests › should navigate to the home page and find search bar (18.7s)
✓ 3 [chromium] › tests\search.spec.ts:17:7 › Search Bar Tests › Positive: should enter a value in the searchbar and find a product (5.3s)
✓ 4 [chromium] › tests\search.spec.ts:31:7 › Search Bar Tests › Negative: should enter a value in the searchbar and not find a product (4.2s)

4 passed (40.6s)
```

Status: All tests Passed